How To Configure AWS Certificate Manager For SSL/TLS

- We all know the reasons we need to configure an SSL certificate on our websites. SSL certificates (sometimes referred to as TLS certificates) help safeguard our data online. An SSL certificate brings forth a robust encryption layer to play. With this encryption, it becomes extremely hard for intruders to intercept data in transit.

  The whole process of purchasing, installing, configuring, and renewing SSL certificates can be somewhat daunting. AWS certificate manager helps eliminate this hassle. It simplifies everything even for the non-tech users. With AWS certificate manager, certificate acquisition is a breeze. Certificate management is centralized and installation, configuration, and renewal are a piece of cake. So you are probably saying to yourself, “well, let me sign up to AWS certificate manager and get things up and running.” Good idea. But you need to have a perfect idea of how AWS certificate manager works.

What Is An SSL Certificate?

SSL (Secure Sockets Layer) is a digital certificate that validates the identity of a website and secures the data transmitted between the user’s browser and the website’s server. It provides a secure, encrypted connection. These encrypted sessions prevent unauthorized access to confidential information. Confidential info may include things such as login credentials, payment details and personal information.

SSL certificates also help build trust and security online. They indicate that the website is trustworthy and uses encryption protocols to ensure data integrity and confidentiality.

At this point, it is crystal clear that an SSL certificate is an indispensable ingredient to any website that intends to survive the current wave of cyber threats. There are several SSL certificate options available. Just ensure you get yourself one that fits your needs.

SSL resellers such as CheapSSLShop offer valuable and budget friendly certificate options that will do just good for you.

What Is AWS Certificate Manager (ACM)?

AWS Certificate Manager (ACM) is a service that streamlines the process of acquiring, managing, and deployment of both public and private SSL/TLS certificates to be used with AWS services and internal resources.

ACM simplifies the previously time-consuming and labor-intensive operations associated with purchasing, uploading, and renewing SSL/TLS certificates. They save time and eliminate administrative costs. Through the automation of these procedures, ACM ensures that certificates are always up-to-date and in line with security standards, strengthening the overall security position of AWS deployments.

From securing public-facing websites to internal network connections, ACM offers an intuitive interface and easy integration with other AWS services to ensure strong encryption and authentication mechanisms.

Why To Use AWS Certificate Manager?

Here are the three reasons why we strongly advocate for AWS certificate managers:

- **Simplified SSL/TLS Enablement:** ACM simplifies the process of activating SSL/TLS for websites or applications that are hosted on AWS infrastructure. Manual tasks related to SSL/TLS certificate management are eliminated, which leads to the reduction of complexity and saves time for developers and administrators.
- **Efficient Management of Multiple Domain Names:** ACM simplifies the management of multiple domain names by allowing you to include additional fully qualified domain names (FQDNs) in a single certificate. In other words, you get the multi-domain benefits by signing up to AWS certificate manager. With ACM, you're

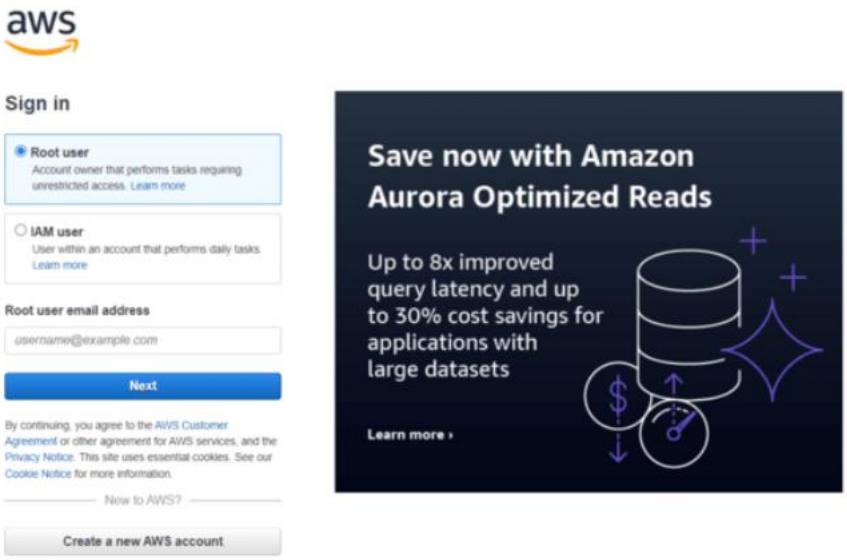
no longer faced with the dilemma of choosing between acquiring a wildcard SSL or a cheap multi-domain certificate, as ACM offers an efficient and cost-effective solution for securing all your domains under one umbrella.

- **Automated Certificate Management**: ACM takes care of certificate renewals which help to prevent downtime due to improperly configured, revoked, or expired certificates. ACM's automation of these processes guarantees that certificates are always current and meet the required security standards, thereby improving the trustworthiness and security of the applications.
- **Centralized Certificate Management:** With ACM, all SSL/TLS certificates in an AWS Region can be centrally managed via the AWS Management Console, CLI, or APIs. This centralized approach makes it easier to manage certificates, provides better visibility, and standardizes security practices throughout your AWS infrastructure. Moreover, ACM ensures protection of certificate private keys with strong encryption and key management best practices that promote security and compliance.

Step By Step Guide on How to Configure AWS Manager for SSL/TLS

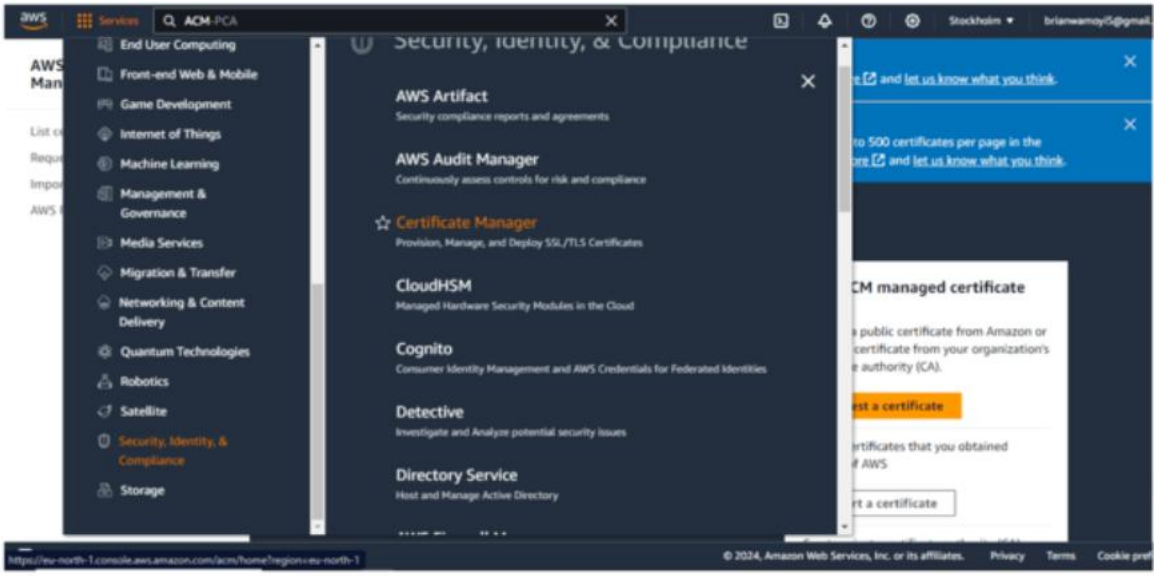
Step 1: Sign In To AWS Certificate Manager Console

- Visit the AWS Management Console and sign in to your AWS account using your credentials.
- If you don't have an account, you can create one by following the instructions provided. See the screenshot below:



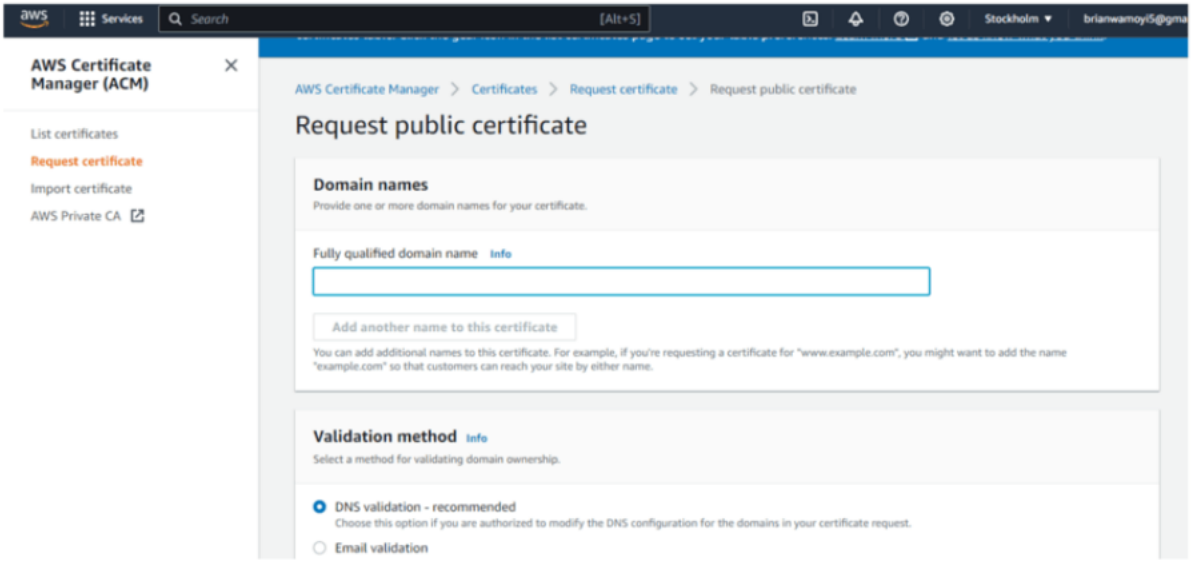
Step 2: Navigate to ACM

Once logged in, navigate to the AWS Certificate Manager (ACM) service by typing "ACM" in the search bar or locating it under the "Security, Identity, & Compliance" section.



Step 3: Enter a custom domain name for your API

- In the ACM console, click on the "Request a certificate" button.
- Enter the custom domain name for which you want to configure SSL/TLS encryption.
- You can also specify additional domain names (subject alternative names) if required.



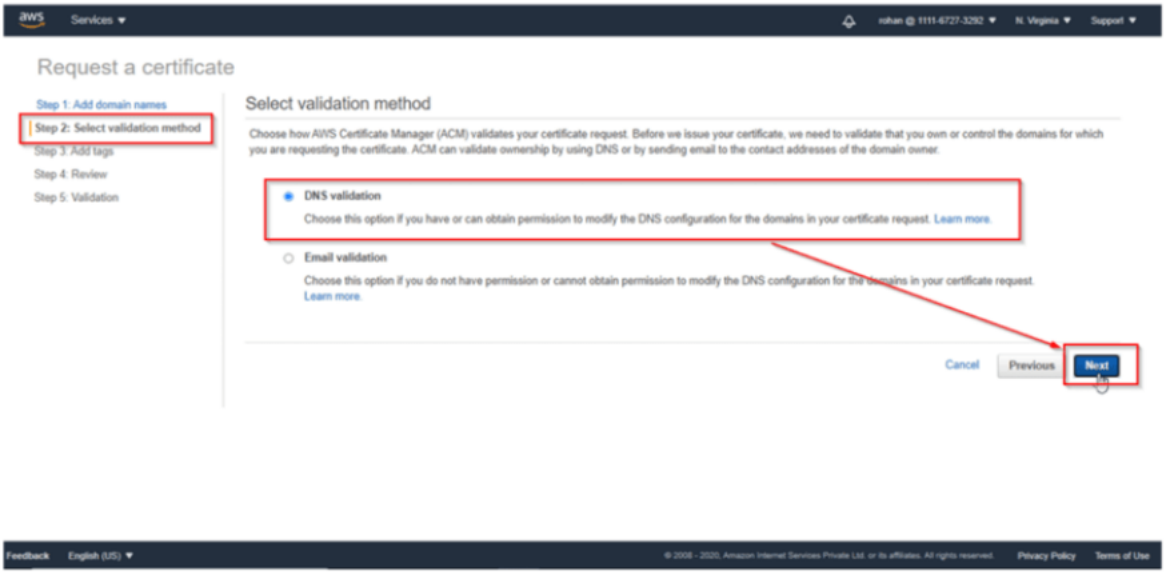
Step 4: Review And Confirm

- Review the details you've entered for the certificate request and ensure they are accurate.
- Confirm the certificate request to proceed to the next step.

Step 5: Domain Validation

- Choose the validation method for your certificate, either email validation or DNS validation.
- If you choose email validation, follow the instructions in the email sent to the domain owner's email addresses to validate domain ownership. If you

choose DNS validation, add the provided DNS records to your domain's DNS configuration.



Step 6: Certificate Issuance

- Once domain validation is complete, ACM will issue the SSL/TLS certificate for the specified domain name(s).

Step 7: Using The Certificate

- Integrate the issued certificate with your AWS resources requiring SSL/TLS encryption, such as EC2 instances, Elastic Beanstalk environments, or any other service supported by ACM.
- Monitor the certificate's status and manage it through the ACM console as needed. ACM will automatically handle certificate renewal before expiration.